

ASSIGNMENT - USB DRIVE ACQUISITION

OVERVIEW

You are back at the digital forensics lab and you need to begin the process of acquiring a forensic image of the seized USB drive. For this assignment you will be required to travel to the UTSA forensics lab with the previous assignment's USB drive. Once at the lab, you will use your laptop and write-blocker to obtain a forensic image of the USB drive. This process may take up to an hour, so plan accordingly. The location of the lab and hours will be provided in the LMS.

INSTRUCTIONS



Case Number: YY-T003



You will be using the IoT forensic lab's evidence imager. **All evidence will be stored in the D:folder!**

Phase One

1. Sign up for using accessing the UTSA IoT forensics lab
2. Access the lab and imager workstation
3. Sign in **BEFORE ACCESSING THE SYSTEM**
4. Before connecting to/from the write-blocker
 - o Connect to power
 - o Power on the device
 - o Power off the device
5. Connect the evidence device to the write-blocker
6. Power on the write-blocker and wait for the evidence device to mount to the HOST system
7. Using Powershell in your Windows virtual machine, obtain
 - o Disk information
 - o Partition information
 - o Volume/filesystem information
 - o Serial number of the device

Phase Two

1. Login to imager workstation
2. Create a folder with your last name in the **D:\Evidence** folder
 - o This is where your evidence should be stored
3. Open FTK Imager
4. Begin imaging the device under these requirements
 - o Raw (dd) image
 - o Name the file "usb-01-image"
 - o Set "0" for the "Image Fragment Size (MB)"
5. Wait for the process to complete
6. Ensure the verification process completes and the result is "MATCH"
 - o Otherwise you will have to reimage the device
7. Review the usb-01-image.001.txt file
8. Secure your evidence

During each phase

1. Document all procedures in your Rocketbook
2. Document tool outputs
 - o Screenshots!
3. Gather as much information possible BEFORE leaving the lab

QUESTIONS

1. What is the disk information for the evidence device?
2. What is the partition information for the evidence device?
3. What is the volume information for the evidence device?
4. What is the file system information for the evidence device?
5. What is the serial number of the device?
6. When did imaging begin?
7. When did imaging end?
8. Was the process successful?
9. Was the image verified to the evidence device? (check the imaging log)
10. What is the overall size of the forensic image in bytes?
11. What is the MD5 and SHA1 hash of the forensic image?
12. What is the MD5 and SHA1 hash of the evidence device?

EVIDENCE

Name	Description
USB Drive	Amazon Basics 128 GB USB drive (black)

REQUIRED TOOLS

Name	Vendor
FTK Imager	Enterro / Access Data
Hardware write-blocker	Tableau
Laptop	-

REQUIRED FILES

- evidence-report-template.docx

DELIVERABLE REQUIREMENTS

- Evidence report with the following sections and requirements
 - Executive Summary
 - Evidence Seized
 - Tools Used
 - Case Information, Observables, Investigator Processes
 - Question answers
 - Conclusion
 - Exhibits (Not required)
 - References (Not required)
 - Photos/screenshots, w/ captions, are required to justify your findings
 - In line with your question answers, not in an appendix/exhibit
 - Must use report template

GRADING RUBRIC

- Evidence report - 100 pts
 - Completed answers - 60 pts
 - Answered questions for each item - 2.5 pts
 - Provided screenshot for each item - 2.5 pts

- Used template - 30 pts
- Following instructions - 10 pts